

Roll No.

--	--	--	--	--	--	--	--	--	--	--	--

Total No. of Pages : 02

Total No. of Questions : 09

MCA (2015 & Onward) (Sem.-5)
Artificial Intelligence
Subject Code : MCA-501
Paper ID : [74381]

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTIONS-A, B, C & D contains TWO questions each carrying TEN marks each and students has to attempt any ONE question from each SECTION.
2. SECTION-E is COMPULSORY consisting of TEN questions carrying TWENTY marks in all.

SECTION-A

1. a) Define AI. Discuss the history of AI in brief.
b) What is an AI technique? Give some examples of problems to explain AI techniques.
2. Can production systems be described by a set of characteristics? If so, what relationships are there between problem types and the types of production systems best suited to solving the problems?

SECTION-B

3. What is A* search? Explain various stages of A* search with an example.
4. Explain Alpha-beta pruning with algorithm and an example.

SECTION-C

5. What are the limitations of propositional logic to represent the knowledge base? Briefly discuss the syntax of FOPL.
6. What is Resolution principle? Discuss the algorithm for propositional resolution with an example.

SECTION-D

7. What is an Expert system? Discuss the components and architecture of an expert system.
8. Define natural language processing. Discuss the phases in processing a natural language sentence.

SECTION-E

9. Briefly answer the following :

- a) Define Problem Space.
- b) What is the role of Turing test in AI?
- c) Write the state space search for the Water Jug problem.
- d) Define Unification.
- e) What is Modus Ponens rule? Give an example to show how it works.
- f) What is default reasoning?
- g) What is the difference between backward and forward chaining?
- h) What is clausal form?
- i) Write the general form of Bayes' theorem.
- j) What is Parsing?